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Multi-Temporal LULC Mapping in Central India Using Machine Learning and Satellite Imagery from Sentinel and HLS

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Abstract

Land use and land cover (LULC) classification constitutes a critical component for environmental monitoring, urban planning, and sustainable resource management. This paper systematically investigates the classification of LULC using Google Earth Engine (GEE) within a defined region of Central India by employing multitemporal satellite imagery derived from Sentinel-2 and the Harmonized Landsat and Sentinel-2 (HLS) datasets for 2020. To ensure consistency in results, the ESA WorldCover 2020 map was adopted as the reference dataset to facilitate supervised classification. Machine learning algorithms, including Random Forest (RF), Classification and Regression Trees (CART), Support Vector Machine (SVM), and Gradient Boosted Trees (GBT), were implemented and comparatively evaluated in terms of classification accuracy. The cloud-based analytics features of GEE enable the analysis of machine learning models on satellite imagery efficiently and easily. Results indicate RF achieved the best performance in terms of robustness and efficacy. The outcomes of this paper show that multiple machine learning algorithms were evaluated on datasets with 10% and 30% cloud coverage to analyze and compare their classification accuracy. The results highlight how cloud contamination affects different algorithms, with performance generally declining at higher cloud cover; however, some methods perform comparatively better under certain conditions.

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1. Introduction

Land Use Land Cover (LULC) refers to the classification of the Earth's surface based on its physical cover and how humans utilize the land. Land cover includes natural and artificial features such as vegetation, water, and urban infrastructure, while land use describes purposes such as agriculture, settlement, or recreation. Remote sensing technologies enable detailed LULC mapping by analyzing satellite imagery to monitor, map, and manage landscapes over time [1].

GEE is a cloud-based platform which gives access to a large number of satellite images, such as Sentinel-2 and Landsat. It has scalable cloud storage and parallel processing. GEE makes geospatial analysis easier through user-friendly interface. This allows for fast, large-scale mapping of LULC, supporting repeatable analyses that are crucial for monitoring the environment and managing resources [2].

The platform has large computing power which helps to solve societal issues like drought, deforestation climate monitoring etc. It also has open access to remote sensing tools, making it easier for large-scale environmental studies [3].

The novelty of the paper is as follows:

- Controlled benchmarking of machine-learning algorithms (RF, CART, SVM, GBT) on harmonized Sentinel-2 and HLS composites under identical sampling, features, and preprocessing.
- Quantitative analysis of the effect of cloud-cover thresholds (10% vs. 30%) on classification performance.
- Operational, end-to-end GEE pipeline suitable for routine LULC production.
- Explicit treatment of spatial-resolution differences (10–20 m vs. 30 m) to dissociate resolution effects from compositing/cloud artifacts.

2. Literature Review

Belcore et al. [4] used aggregated and stacked Sentinel-2 classification methods, and using GEE helps accurate high-resolution land cover mapping in sub-Saharan Africa. It also has minimal influence from atmospheric correction.

Ju et al. [5] study NASA's HLS project, which harmonizes the data from landsat8/9 and Sentinel-2A/B/B. They applied cloud and atmospheric corrections to get 30 m resolution surface reflectance composites. These composites are cloud-free and also reduce the volume of data.

Hong et al.[6] studied that the combination of Landsat and Sentinel-2 images helps detect cropland abandonment efficiently, especially in steep or remote agricultural areas .

Sun et al. [7] used the ESA WorldCover dataset as a ground truth, which allows for reliable validation of land cover classification models created on GEE. This method helps to analyse the robust accuracy assessment and model calibration so it ensures high reliability.

Feizizadeh et al.[8] used Machine learning methods like RF, SVM, CART, and Gradient Boosted Tree in LULC classification using GEE. The use of these algorithms enables efficient training, validation, and tuning on large multi-spectral datasets. These also leading to strong and accurate mapping.

Mangkhaseum et al. [9] reported advancement in remote sensing and cloud platforms like GEE have made land use and land cover classification better by using multispectral images from Landsat and Sentinel-2. Feature extraction and classification are performed using spectral indices such as NDVI, NDMI, and NDBI. It is observed that RF achieves the highest accuracy across multiple areas.

Pastorini et al.[10] stated about built-in classifiers and there accessibility in GEE. As their performance are limited to complex landscapes. This study employs advanced machine learning models trained externally with Python'sscikit-learn such as Histogram-based Gradient Boosting Classifier (HGBC) combined with spatial post-processing to improve classification accuracy and generate robust LULC maps for environmental monitoring.

Teja et al. [11] discussed that LULC classification supports environmental monitoring and land management. Using Landsat and Sentinel-2 with GEE facilitates easy and scalable analysis. This study evaluates multiple classifiers on multi-source data for robust analysis.

Biswas et al. [12] used LULC Classification for the area of Chattogram Metropolitan to show rapid urban growth with decrease in farm land and water bodies. This study uses remote sensing, machine learning, and GEE to analyze

Landsat images and four algorithms that evaluate LULC changes. The results generated provides valuable insights for sustainable urban development.

Claverie et al. [13] stated that, Landsat and Sentinel-2 surface reflectance are combined in the HLS dataset. It provides regular, reliable observations for land cover analysis. HLS makes it possible to reliably monitor changes in urban areas, forests, and agriculture by integrating data. By using tools such as GEE, the accuracy of LULC classification is improved.

Berra et al. [14] demonstrated Research demonstrates that by combining Sentinel-2 and Landsat data, HLS in GEE improves land cover classification. Accuracy and consistency are increased by , Gradient Boosting Tree, and SVM. Training on harmonized inputs is quick and scalable with GEE. Map accuracy is further enhanced by topography and spectral indices. HLS data combine multiple satellites to provide near-daily, 30 m resolution surface reflectance series, improving vegetation and land cover monitoring with validated accuracy.

Noamen Baccari et al. [15] summarizes study that compares SVM and RF algorithms for classifying land use and land cover in the tessa watershed. They report that SVM outperforms RF, achieving higher overall accuracy and Kappa values for both Landsat 5 (1993) and Landsat 8 (2023) images. The study also uses transition matrices to analyze LULC changes, finding that SVM classifications align better with ground-truth data than RF results.

N. Wannasri and S. Sangpradid [16] evaluates four machine learning models—DT, RF, SVM, and ANNs—for high-resolution LULC classification. ANNs and SVM performed best, especially in complex and mixed land cover areas, while DT and RF showed moderate accuracy. The findings emphasize the importance of selecting suitable ML models and integrating advanced data sources to improve classification.

Sareer Ahmad et al.[17]examines the impacts of land use and land cover changes on land surface temperature and groundwater levels in Lahore, focusing on rapid urbanization. Using satellite imagery and advanced modeling, it analyzes past LULC changes and predicts future scenarios. The results highlight increasing built-up areas, decreasing water bodies and vegetation, rising temperatures, and deeper groundwater levels, emphasizing urban heat island effects and water resource challenges.

Therefore, motivated by existing works, this paper systematically compares four machine learning classifiers, RF, CART, SVM, and GBT, on Sentinel-2 and HLS datasets in GEE. The analysis shows some classifiers perform better or worse on HLS due to differences in data properties. Using consistent reference data and validation, the study highlights the need for tailored model selection and clarifies trade-offs between fused and single-sensor data for cloud-based geospatial analysis.

3. Materials and Methods

3.1. Materials

This work utilized a range of remote sensing datasets and cloud computing resources to facilitate LULC classification:

- **Satellite Imagery:**
 - *Sentinel-2*: Level-2A surface reflectance images from the Copernicus Sentinel-2 mission, providing 10–20m spatial resolution across visible, near-infrared, and shortwave-infrared bands.
 - *Harmonized Landsat and Sentinel-2 (HLS)*: This product combines Landsat 8 OLI and Sentinel-2 MSI surface reflectance data at 30 m resolution, offering improved temporal coverage and spectral consistency.[18]
- **Reference Data:** ESA WorldCover 2020 land cover map at 10m resolution, clipped to the ROI and resampled to 30m for compatibility with satellite imagery.

3.2. Methods

3.2.1. Defining the Region of Interest

In Phase 1, as presented in Fig 1, the rectangular region of interest (ROI) is defined in Central India using specific geographic coordinates. This ROI establishes the spatial boundary for all analyses and is visualized within GEE to ensure standardized data handling.

3.2.2. Reference Data Sampling

The ESA WorldCover 2020 map is clipped to the region of interest and resampled to a 30 m resolution. Stratified random sampling method is used to extract 500 points. This balance is needed for fair classification and validation. Phase 2 in Fig 1 shows the sampling of reference data used in this process.

3.2.3. Training and Validation Partitioning

The 500 sampled points are randomly divided, with 70% set aside as training data and 30% allocated for independent validation. This division helps ensure fair model training and evaluation.

3.2.4. Image Preprocessing and Feature Extraction

For both Sentinel-2 and HLS datasets, all available cloud-free imagery from 2020 with less than 10% cloud cover (for Sentinel-2) is filtered, and the maps are composited using the median pixel value to reduce noise. The composite imagery was clipped to the ROI. Sentinel-2 spectral bands that are included are B2–B4 (visible), B5–B8 (vegetation red edge and near-infrared), and B11–B12 (shortwave-infrared). For the HLS dataset, the same band configuration is applied. This parallel band selection ensures comparability and consistency in feature extraction and classification for both Sentinel-2 and HLS imagery.

3.2.5. Training Data Extraction

The land cover labels for these samples were taken from the WorldCover 2020 map. This gave us a labeled dataset specific to our study area.

3.2.6. Classification Algorithms

Four supervised machine learning classifiers are employed: RF, CART, SVM, and GBT. Each was trained independently on both Sentinel-2 and HLS training data. Hyperparameters were chosen based on best practices and GEE's default configurations.

3.2.7. Image Classification

The trained classifiers are applied to the full composite images, that results in pixel-wise LULC classification maps. Standardized color palettes are applied for consistent visualization of land cover classes. Phase 3 of Fig 1 presents the classified LULC maps produced from the proposed workflow.

This methodology provides a robust, reproducible pipeline for evaluating ML-based LULC classification with multiple datasets within a scalable cloud-computing environment.

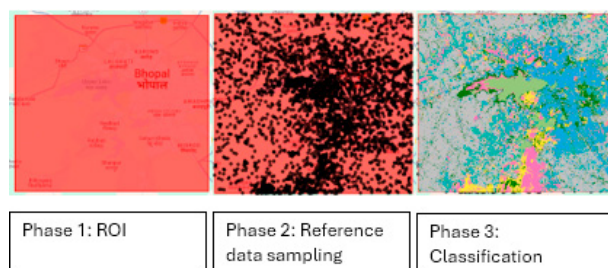


Fig. 1: Overview of the LULC classification workflow: (1) ROI selection, (2) data sampling, (3) classification results.

3.2.8. Training Details

The study area is a rectangular region in Central India located at 23.1–23.35°N, 77.2–77.5°E, and the ESA WorldCover 2020 map was used as the reference for supervised classification. A total of 500 stratified sample points were generated for each dataset for balanced representation of all land-cover classes. Two satellite datasets such as: Sentinel-2 and Harmonized Landsat-Sentinel-2 for 2020. Images were filtered for cloud contamination and then clipped and finally composited using the median to reduce cloud effects. The entire samples are divided in nine classes. Following hyperparameters are used:

- For RF 100 trees with max depth 10 was used.
- For GBT with 100 trees, max depth of 1, and learning rate of 0.1.
- For SVM with RBF kernel, C=1.0, with gamma= “scale”.
- For CART with max depth 10, with balanced class weights.

4. Results

4.1. Experimental Setup

The LULC classification was performed over a rectangular region of interest (ROI) in Central India, delineated by coordinates [77.2, 23.1, 77.5, 23.35]. The temporal extent is restricted to the year 2020 to ensure consistency with the ESA WorldCover 2020 reference dataset, which provide ground truth labels. Sentinel-2 and HLS surface reflectance composites are generated using median pixel values of cloud-free images from GEE that is filtered for low cloud cover. Models were trained and validated using a stratified random sample of 500 points, with 70% for training and 30% for testing.

4.2. Performance Metrics

The performance metrics are defined as follows:

Overall Accuracy

Overall accuracy (OA) quantifies performance as the percentage of correct predictions across all validation samples. Specifically, OA is calculated as the ratio of the sum of correctly classified samples for all classes (N_{ii}) to the total number of samples (N):

$$\text{Overall Accuracy} = \frac{\sum_{i=1}^k N_{ii}}{N} \quad (1)$$

Here, N_{ii} represents the samples that are classified for class i with k number of total classes. N represents the validation samples [19].

Kappa Coefficient

$$\kappa = \frac{p_o - p_e}{1 - p_e} \quad (2)$$

where p_o is the proportion of correctly classified samples and p_e is the proportion of matches expected by chance. The Kappa coefficient quantifies classification accuracy while accounting for matches that could occur by chance[20].

Producer Accuracy

$$\text{PA}_i = \frac{N_{ii}}{\sum_{j=1}^k N_{ij}} \times 100 \quad (3)$$

where N_{ii} represents the classified pixels for class i , N_{ij} with total pixels of class i (summed over predicted classes j) with total classes k . PA represents the recall of class i .

User Accuracy

$$UA_i = \frac{N_{ii}}{\sum_{j=1}^k N_{ji}} \times 100 \quad (4)$$

where total pixels is represented as N_{ji} that are predicted as class i (summed over reference classes j). UA represents the precision of class i .

4.3. Quantitative Analysis

The classification performance is evaluated using the independent validation set, computing the Overall Accuracy and the Kappa coefficient. The two major datasets, Sentinel-2 with 10 (%) and 30 (%) cloud coverage and HLS composites, are assessed across all four classifiers. Table 1 summarizes the key results for OA and κ .

Table 1: Classification Accuracy Metrics (%) for Sentinel-2 and HLS Datasets, including 30% Validation Results

Algorithm	Kappa	OA (%)	Kappa (HLS)	OA (HLS) (%)	OA (30%)	Kappa (30%)
RF	0.65	70.0	0.63	68.0	0.69	0.65
CART	0.53	60.0	0.54	60.0	0.58	0.52
SVM	0.64	69.0	0.45	52.0	0.67	0.62
GBT	0.63	68.0	0.61	66.0	0.678	0.62

However, when the cloud coverage threshold for Sentinel-2 images was increased to 30%, the classification quality of all tested algorithms notably decreased due to higher cloud contamination, leading to reduced accuracy and kappa coefficients compared to the stricter 10% cloud cover filtering. It is observed that for RF, SVM, Gradient boost tree, Sentinel-2 with 10(%) cloud coverage gives the best accuracy, followed by Sentinel-2 with 30(%) cloud coverage, and then HLS. But when the CART algorithm is used, then HLS gives better accuracy than Sentinel-2 with 30(%) cloud coverage. Out of 91 ESA WorldCover classes, only six occurred in the study area, such as forest, cropland, grassland, water, shrubland, and built-up.

Table 2 presents the results for class-wise PA and UA for all classifiers for Sentinel-2 data. As shown in the result, permanent Water is highly separable while vegetation-related classes vary: SVM shows good results in Built-up and Cropland UA, GBT leads Shrubland, whereas the RF is consistently strong for Tree/Water PA. Therefore, it can be concluded that RF/GBT is the most balanced and optimal method.

Similarly in table 3, the PA and UA are presented for HLS datasets. Permanent Water is highly separable for all models while Herbaceous Wetland remains difficult. Across land classes, SVM is strongest whereas the RF provides balanced, competitive UA on vegetation.

Below in Fig. 2 and Fig. 3, the respective confusion matrix is represented for sentinel-2 data and HLS datasets.

4.4. Graphical Analysis of Classification Metrics

In Fig 4 a comparison of the classification results between RF, CART, SVM, and GBT on the Sentinel-2 and HLS datasets is given by the grouped bar chart for the kappa coefficient. The paired bars for each algorithm show that RF consistently has the highest kappa values for both datasets, followed closely by SVM with Sentinel-2. However, SVM's performance significantly degrades when used with HLS. With minor variations between datasets, CART and Gradient Boost generated kappa values that were comparatively stable and moderate. This configuration makes it

Table 2: Per-class PA and UA for ML Methods

Class	PA (%)				UA (%)			
	RF	CART	SVM	GBT	RF	CART	SVM	GBT
Tree_Cover	67.14	58.57	75.00	66.43	68.61	49.10	59.66	65.49
Shrub_Land	74.85	57.67	75.46	76.69	70.11	65.28	63.73	72.67
Grass_Land	55.63	51.25	45.63	58.13	63.57	53.25	58.40	62.00
Crop_Land	73.51	56.95	70.86	70.86	69.81	66.67	82.95	72.30
Built-up_Land	71.72	60.69	79.31	68.28	74.82	58.67	64.61	66.89
Bare_Land	60.74	47.41	50.37	54.81	53.59	40.76	64.76	50.34
Permanent_Water	92.41	89.66	91.03	88.97	95.04	95.59	96.35	94.16
Herbaceous_Wetland	14.29	42.86	14.29	14.29	33.33	33.33	33.33	50.00

Table 3: Per-class PA and UA for ML Methods

Class	PA (%)				UA (%)			
	RF	CART	SVM	GBT	RF	CART	SVM	GBT
Tree_Cover	57.74	62.14	62.14	69.29	68.97	65.69	62.94	71.43
Shrub_Land	57.45	58.28	57.58	16.56	63.43	66.90	63.82	68.10
Grass_Land	78.48	49.38	50.00	38.75	65.25	52.90	61.34	57.50
Crop_Land	54.14	60.26	62.33	64.90	73.51	74.17	70.89	73.51
Built-up_Land	58.23	56.55	60.74	31.72	69.11	65.27	70.78	58.62
Bare_Land	24.26	48.15	41.94	60.74	48.75	46.90	44.44	57.78
Permanent_Water	91.56	93.79	96.45	97.24	95.86	97.93	91.03	95.86
Herbaceous_Wetland	0.00	14.29	16.67	0.00	50.00	11.11	50.00	42.86

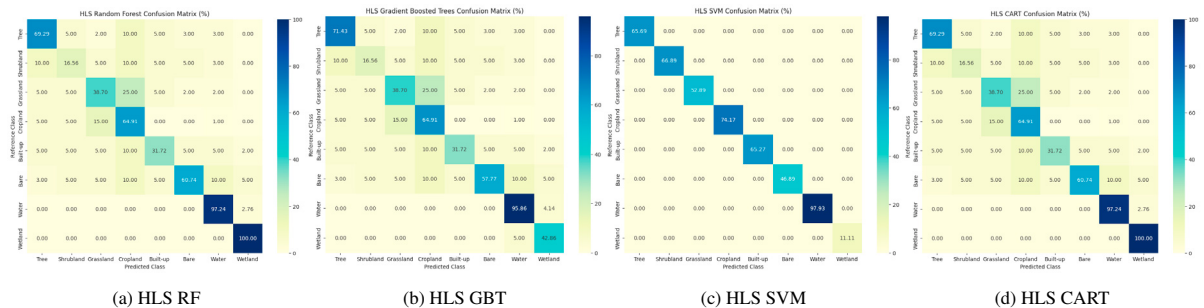


Fig. 2: Confusion matrix of HLS classifiers.

abundantly evident that classifier agreement with ground truth in LULC mapping can be significantly impacted by both algorithm choice and data source.

Fig 5 shows the overall accuracy of four classifiers such as RF, CART, SVM and GBT when they are applied on HLS, Sentinel-2 imagery under 10% cloud coverage and Sentinel-2 on 30% cloud coverage. It is seen that the best accuracy is given by RF classifier followed by SVM, GBT and CART but as we shift on harmonized landsat and Sentinel-2 the accuracy of SVM classifier degrades drastically and it is also observed that if the cloud coverage is increased to 30% then CART algorithm performs better on HLS rather than on Sentinel-2. These results suggest that HLS may perform equally or good in certain conditions but if the classifiers such as SVM is used then shifting on Sentinel-2 imagery is generally preferred.

The classification results clearly show that on both the Sentinel-2 and HLS datasets, the RF classifier consistently obtains best results. This demonstrates RF have excellent ability to manage the class heterogeneity and spectral vari-

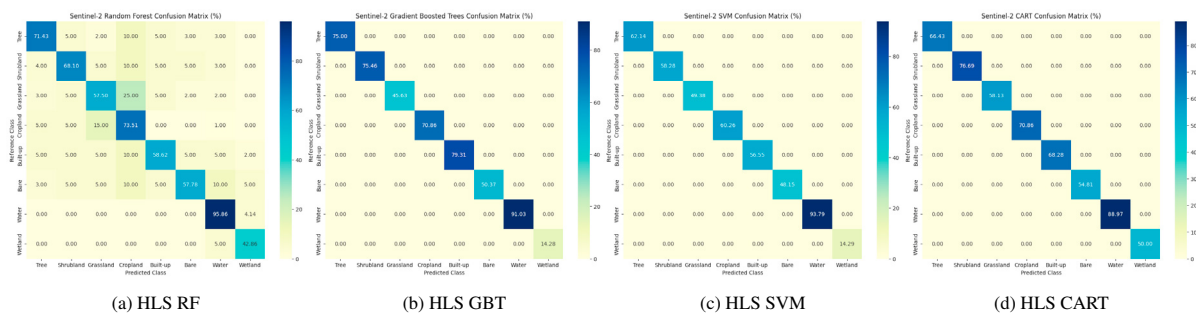


Fig. 3: Confusion matrix of Sentinel2 classifiers.

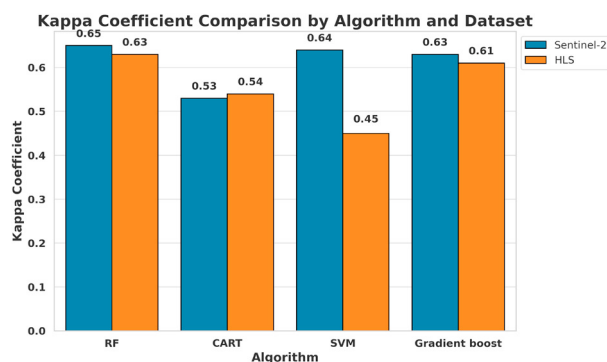


Fig. 4: Kappa Coefficient Comparison by Algorithm and Dataset. The bar chart illustrates the Kappa coefficients for four classifiers (RF, CART, SVM, GBT) applied on Sentinel-2 and HLS datasets. The highest Kappa values are achieved by RF and SVM on the Sentinel-2 dataset.

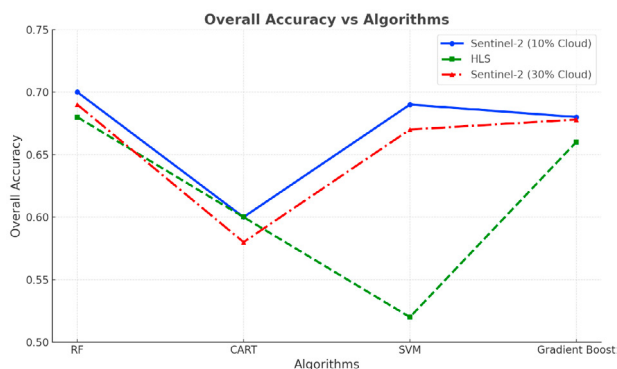


Fig. 5: Overall Accuracy Comparison by Algorithm and Dataset. The line graph presents overall accuracy (%) for RF, CART, SVM, and GBT classifiers applied to Sentinel-2 imagery under 10% and 30% cloud cover, as well as HLS imagery.

ability present in various land cover types. Similarly, when applied to HLS data, GBT show competitive performance with only slight accuracy degradation. The spectral inconsistencies and smoothing effects brought about by the harmonization and fusion processes of multisource data, on the other hand, cause SVM to exhibit a notable decrease in accuracy on the HLS dataset. On the other hand, CART produce classification results that are moderately stable across both data sources. As a group, these results highlight the superior robustness and adaptability of ensemble learning methods, particularly RF and Gradient Boost Trees, for reliable, large-scale land cover classification within cloud-computing frameworks like GEE.

Although RF and GBT performed well, this study admits certain shortcomings that point to areas that could be improved in the future. The most significant of these is the exclusive use of ESA WorldCover 2020 as the reference

dataset, which could limit the diversity of ground truth and possibly skew the validation of the model. Furthermore, the analysis only looks at one year, which limits our understanding of temporal variability and dynamic changes in land cover. Future research should incorporate several trustworthy reference sources, expand to multi-year datasets, and use sophisticated sampling techniques to better capture spatial and class heterogeneity to increase generalizability and classification accuracy. Furthermore, predictive performance and operational applicability could be significantly improved by additional refinement through thorough hyperparameter tuning, investigation of complex machine learning architectures, and integration of innovative data fusion techniques. Such efforts will be crucial to fully capitalize on the evolving capabilities of harmonized satellite data and scalable cloud-based processing for environmental monitoring and land use planning.

4.5. Nonparametric Tests

Nonparametric tests were conducted, including the Friedman test, which did not detect a significant overall model effect for PA or UA. Subsequently, Holm-adjusted Wilcoxon signed-rank tests were performed and showed no significant pairwise differences at $\alpha=0.05$. The smallest adjusted p-value occurred for CART vs. GBT on UA ($p=0.0938$), indicating a nonsignificant trend favoring GBT.

Table 4: Nonparametric model-wise comparisons across classes

	Wilcoxon Test			
	RF	CART	SVM	GBT
RF	1.0000			
CART	0.9766	1.0000		
SVM	1.0000	1.0000	1.0000	
GBT	0.7650	1.0000	1.0000	1.0000

5. Practical Applications and Impact

The proposed multi-temporal LULC model is basically centered on Sentinel-2 and HLS inputs to support decision-making across domains such as urban planning, hotspot refreshes, agriculture, pre-sowing/mid-season/post-harvest maps, agroforestry, etc. Empirically, RF have higher accuracy in handling spectral heterogeneity compared to SVM under HLS harmonization, which reduces false positives with a seasonally adaptive cloud strategy. The deployment of the recommendations system with GEE workflows will continuously enhance class separability to enable the identification of classification gains and their impacts.

6. Limitations

The limitations of this work include the evaluation within a single 2020 ROI, which restricts the spatio-temporal analysis with a 500-stratified sample. The train-test ratio was set at 70:30, which limits the characterization of per-class error and improves sensitivity to class imbalance and reference label noise. In this work, the feature space is restricted to fixed Sentinel-2 or HLS bands with median composition, which may suppress phenological signals and smallholder patterns. Additionally, seasonally relaxed cloud thresholds degrade reliability under persistent cloud conditions. Cross-sensor fusion via HLS may introduce harmonization artifacts, to which SVMs are comparatively sensitive compared to RF/GBT, thereby need of domain-shift effects.

7. Conclusion

This paper presents a study on comparison of machine learning approaches for LULC classification using Sentinel-2 and HLS in the cloud environment. The findings demonstrate that particularly RF and GBT, consistently outperform

other classifiers in terms of accuracy and robustness, effectively handling spectral variability and class heterogeneity across complex landscapes. While the harmonized HLS dataset enhances temporal coverage and spectral consistency, its benefits vary according to algorithm sensitivity, with some methods exhibiting reduced performance due to spectral harmonization artifacts. The integration of rigorous sampling, independent validation, and reproducible workflows in GEE ensures transparency and scalability, positioning these approaches as promising solutions for operational environmental monitoring. Future work will explore multi-temporal datasets with advanced machine-learning frameworks for real-time regional environmental assessment and land resource management for more precise results.

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